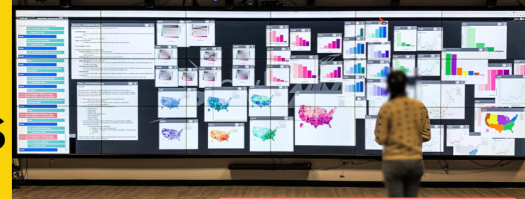


Ditto: Exploring data in large display environments through speech+mid-air pointing interactions



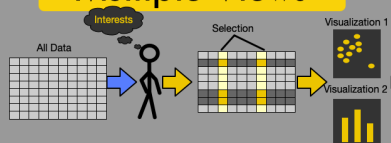
1. University of Cincinnati
2. Taibah University
3. University of Illinois, Chicago
4. Northeastern University
5. University of Hawaii, Manoa

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Data exploration

Multiple Interests, Multiple Views



In data exploration, analysts may identify many points of interest- selections of data values and data attributes. These selections are meaningful partitions of the data and can be represented in multiple views.

Evolving Interests, Incremental Steps



Data exploration is also an iterative process, where analysts move incrementally from one sets of interests to others- new selections of data values and attributes, with multiple rounds of visualization creation.

Large displays

Space for Multiple Views



Large display environments may support visual data exploration by providing abundant space for displaying and juxtaposing the multiple views produced during the exploratory process.

Interaction challenges

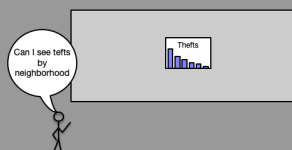
Interaction in large display environments presents challenges. Users should be able to interact with the display at multiple distances, maintain mobility, and interact in collaborative scenarios. Speaking and pointing could fit these requirements.

But- how to achieve this in a way that accommodates large displays showing many views of the data.

Design

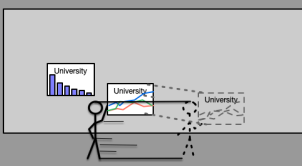
Speak: Rapid View Creation, Supporting Many Points of Interest

First, users of our technique can speak to pose direct requests around their interests (data values and/or data attributes).



Large displays can present many views, our spoken interaction system should help them leverage the space. Users express multiple data value or attribute interests in one command to create a coherent collection of views around these interests.

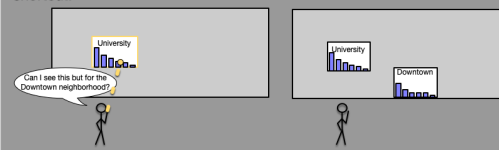
Touch: Position for Sensemaking



Users can freely position these views on the large display, using touch interactions, into configurations that support their exploration process. Such as grouping multiple views showing a common attribute (Eg. year) with distinct filter criteria (Theft in Jan, Theft in May).

Point&Speak: Iterative Exploration

Users can indicate a prior visualization (through pointing) and express that they want to see this view of the data but with some change (through speech), such as a new filter criteria or a new data attribute. Thus, instead of re-articulating a complex request, users can use prior visualizations as a shortcut.

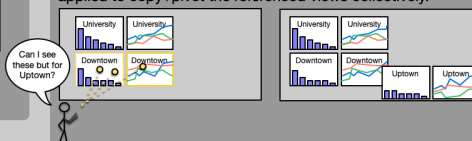


But what if they point to more than one view?

Pointing to more than one view at a time

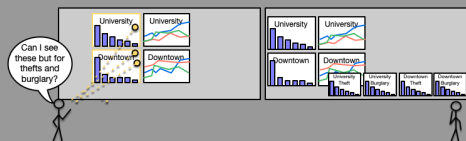
Collective "Copy+Pivots"

The 'pivot point' for this transition can be a common feature shared by the referenced views (such as a common filter criteria or data attribute). The exploratory transition can be applied to copy+pivot the referenced views collectively.



Multiple, Parallel Copy+Pivots

Or users can express multiple pivots (multiple new interests) to apply in parallel to the referenced view or set of views.



Implementation

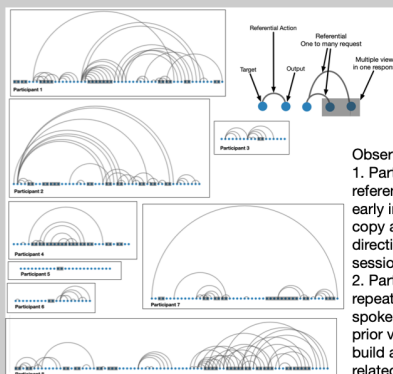


In our implemented system, Ditto, we capture spoken requests through a speech-to-text interface on a phone. A tracking system from a Kinect mounted to the center of the display is interpreted to detect pointing gestures.

Our command interpreter translates speech or speech+pointing inputs into one or several VegaLite specification objects. Our interpreter uses a simple command grammar, which primarily captures user interests (Eg. data values and data attributes and other keywords).

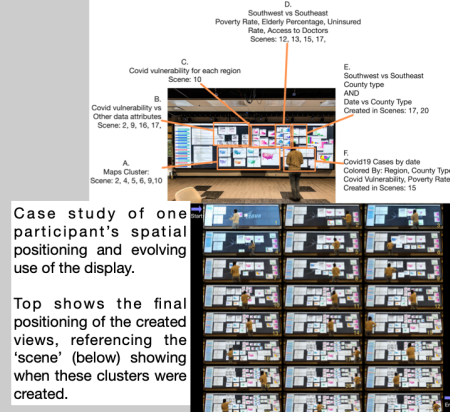
Findings

We evaluated Ditto with 8 participants (plus 4 participants in a pilot study), who explored a COVID-19 dataset. Participants were provided with training, given a detailed data description and exploration prompt. We captured a total of 307 visualization queries with responses from Ditto.



Observations:
1. Participants referenced views from early in their session, to copy and pivot in new directions later in the session.
2. Participants often repeatedly spoke+pointed to a prior view or views to build a collection of related views.

A visualization of participant actions. Blue circles are visualizations in the order they were created. Arcs connect visualizations that were copied and pivoted with speaking+pointing. Views that were created together in one action are shown with a gray box.



Case study of one participant's spatial positioning and evolving use of the display.

Top shows the final positioning of the created views, referencing the 'scene' (below) showing when these clusters were created.